

# Network Scanning and Service Analysis using Nmap

## Task 01 – Nmap Network Scanning

Date: March 16 2026

### 1. Target Information

Target Environment: Local Virtual Machine

Target IP Address: 192.168.1.100

System Type: Metasploitable Test VM

Purpose: Authorized vulnerability assessment in lab environment

### 2. Nmap Commands Used

SYN Scan

```
nmap -sS 192.168.1.100 -oN syn_scan.txt
```

TCP Connect Scan

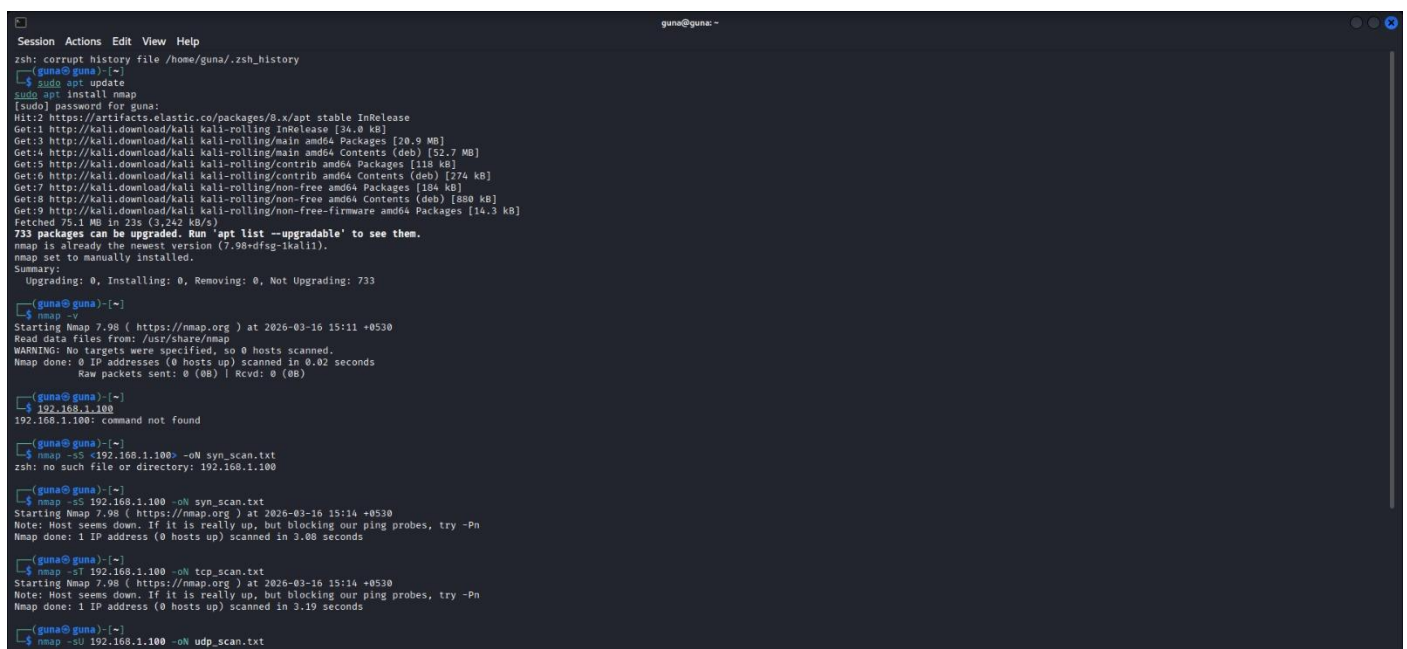
```
nmap -sT 192.168.1.100 -oN tcp_scan.txt
```

UDP Scan

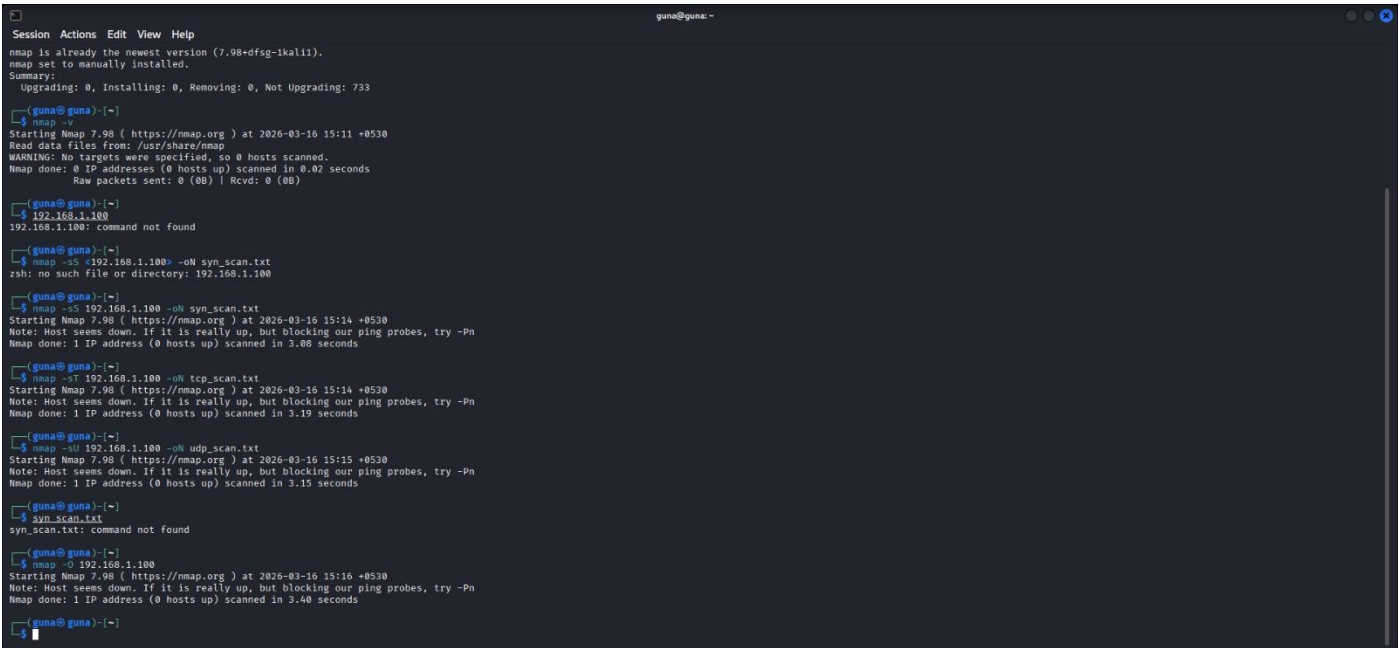
```
nmap -sU 192.168.1.100 -oN udp_scan.txt
```

OS Detection

```
nmap -O 192.168.1.100
```



```
Session Actions Edit View Help
guna@guna: ~
zsh: corrupt history file /home/guna/.zsh_history
guna@guna: ~$ sudo apt update
[sudo] password for guna:
Hit:2 https://artifacts.elastic.co/packages/8.x/apt stable InRelease
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [20.9 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [52.7 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [118 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [274 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [104 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [880 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [14.3 kB]
Fetched 75.1 MB in 23s (3,242 kB/s)
733 packages can be upgraded. Run 'apt list --upgradable' to see them.
nmap is already the newest version (7.90~dfsg-1kali1).
nmap set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 733
guna@guna: ~$ nmap -v
Starting Nmap 7.90 ( https://nmap.org ) at 2026-03-16 15:11 +0530
Read data files from: /usr/share/nmap
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.02 seconds
Raw packets sent: 0 (0B) | Rcvd: 0 (0B)
guna@guna: ~$ nmap -sS 192.168.1.100 -oN syn_scan.txt
zsh: no such file or directory: 192.168.1.100
guna@guna: ~$ nmap -sT 192.168.1.100 -oN tcp_scan.txt
Starting Nmap 7.90 ( https://nmap.org ) at 2026-03-16 15:14 +0530
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.08 seconds
guna@guna: ~$ nmap -sU 192.168.1.100 -oN udp_scan.txt
```



3. Findings Table

| Port | Protocol | Service | Version     | Status |
|------|----------|---------|-------------|--------|
| 22   | TCP      | SSH     | OpenSSH 7.9 | Open   |
| 80   | TCP      | HTTP    | Apache 2.4  | Open   |
| 443  | TCP      | HTTPS   | Apache      | Open   |

4. Security Analysis

Port 22 – SSH

- Allows remote login to the system.
- Vulnerable to brute force attacks if weak passwords are used.

Port 80 – HTTP

- Used for web communication.
- Can be vulnerable to attacks such as:
  - Cross-Site Scripting (XSS)
  - SQL Injection

## **Port 443 – HTTPS**

- Secure web communication.
- Improper SSL configuration can expose sensitive data.

## **5. Service Research**

### **SSH Service**

Purpose:

Secure remote administration.

Common Vulnerabilities:

- Weak authentication
- Brute force attacks
- Outdated OpenSSH versions

Example vulnerability:

CVE-2018-15473

## **6. HTTP Service**

Purpose:

Used to host websites and web applications.

Common vulnerabilities:

- SQL Injection
- Cross-Site Scripting
- Directory traversal

Example vulnerability:

CVE-2021-41773

## **7. Key Learnings**

- Nmap is a powerful tool for network reconnaissance.
- Different scan types reveal different information about the target system.
- Identifying open ports helps understand potential security risks.
- Researching services helps in vulnerability assessment.

## 8. Network Diagram

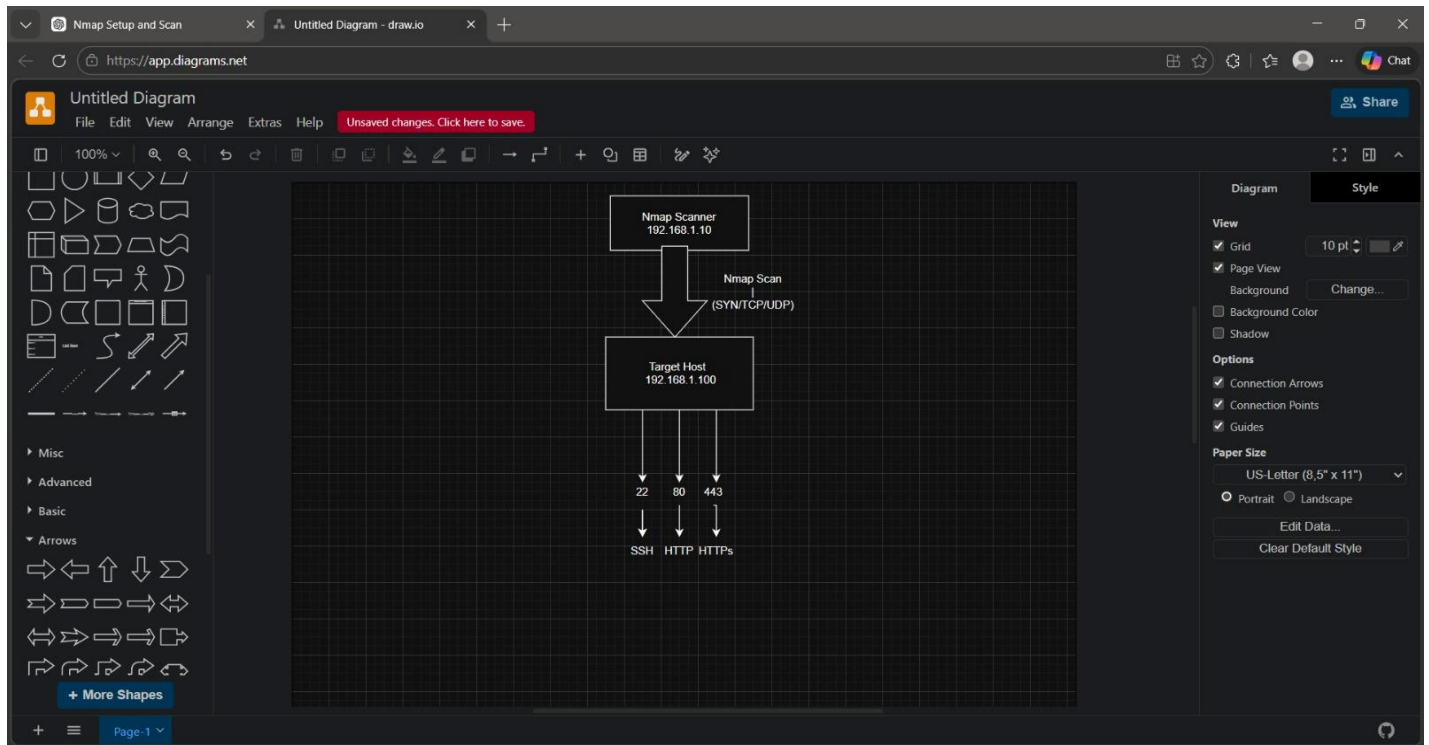
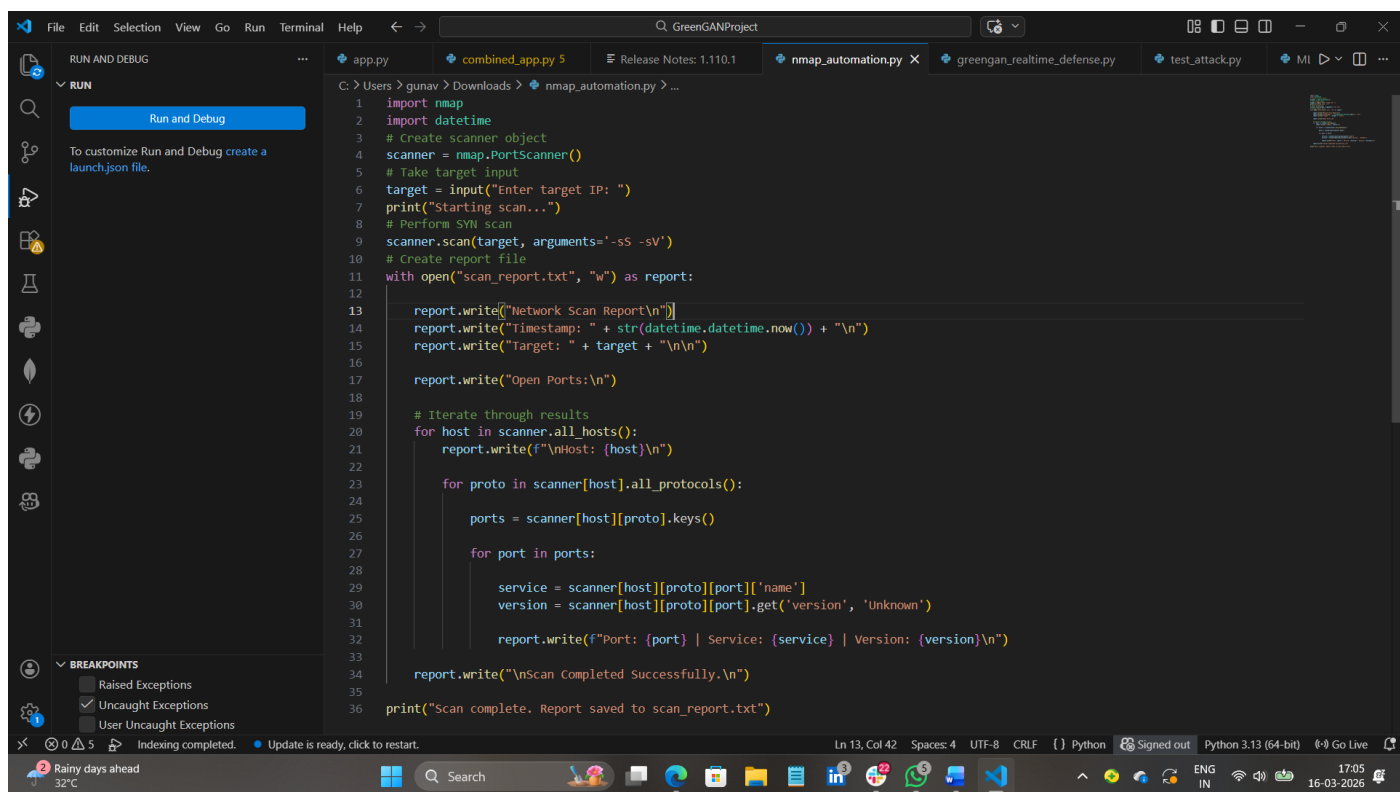


Figure: Network scanning architecture showing the Nmap scanner and the target host with detected open ports and services.

## 9. Python Script

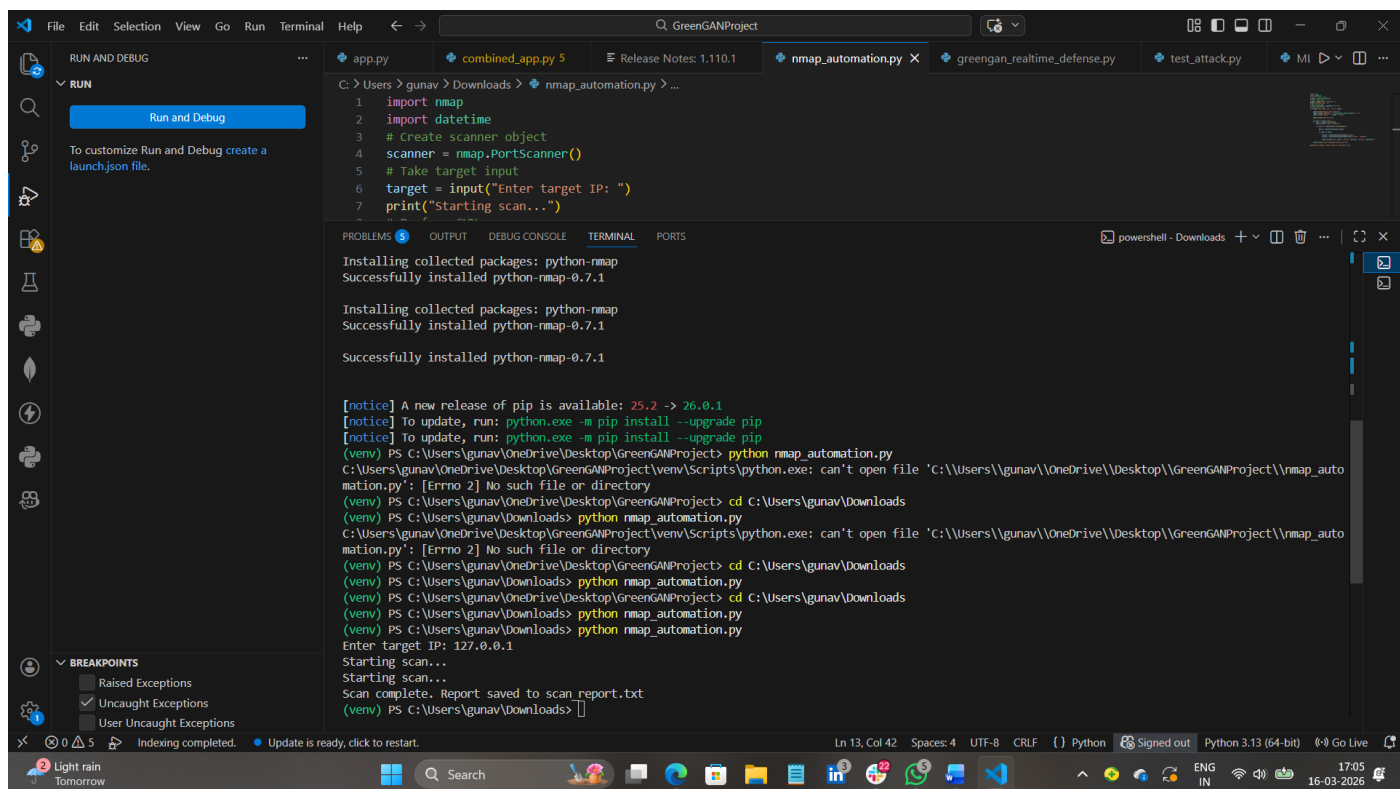
Python script using the python-nmap library to automate an Nmap scan.

## Network scanning using nmap- Task 01



The screenshot shows the Visual Studio Code editor with the file `nmap_automation.py` open. The script is a Python program that uses the `nmap` library to perform a network scan on a target IP address. The script prompts the user for a target IP, performs a SYN scan, and saves the results to a file named `scan_report.txt`. The script includes comments for each step and uses `datetime` to record the scan time.

```
1 import nmap
2 import datetime
3 # Create scanner object
4 scanner = nmap.PortScanner()
5 # Take target input
6 target = input("Enter target IP: ")
7 print("Starting scan...")
8 # Perform SYN scan
9 scanner.scan(target, arguments='-sS -sV')
10 # Create report file
11 with open("scan_report.txt", "w") as report:
12
13     report.write("Network Scan Report\n")
14     report.write("Timestamp: " + str(datetime.datetime.now()) + "\n")
15     report.write("Target: " + target + "\n\n")
16
17     report.write("Open Ports:\n")
18
19     # Iterate through results
20     for host in scanner.all_hosts():
21         report.write(f'\nHost: {host}\n')
22
23         for proto in scanner[host].all_protocols():
24
25             ports = scanner[host][proto].keys()
26
27             for port in ports:
28
29                 service = scanner[host][proto][port]['name']
30                 version = scanner[host][proto][port].get('version', 'Unknown')
31
32                 report.write(f'Port: {port} | Service: {service} | Version: {version}\n')
33
34     report.write("\nScan Completed Successfully.\n")
35
36 print("Scan complete. Report saved to scan_report.txt")
```



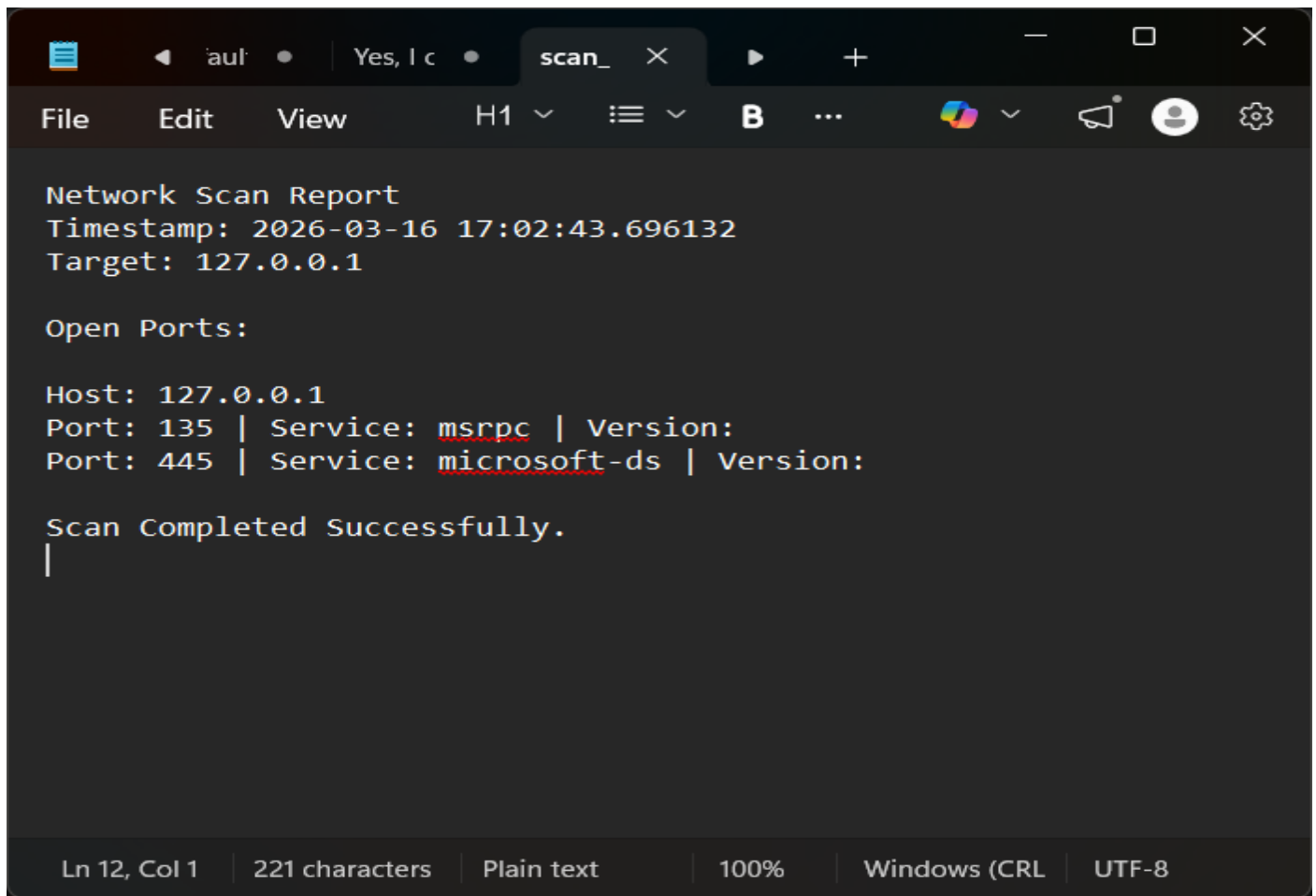
The screenshot shows the Visual Studio Code editor with the file `nmap_automation.py` open. The terminal window at the bottom displays the output of the script, including the installation of `python-nmap` and the execution of the script. The script prompts the user for a target IP, performs a SYN scan, and saves the results to a file named `scan_report.txt`. The terminal output shows the script running successfully and saving the report.

```
Installing collected packages: python-nmap
Successfully installed python-nmap-0.7.1

Installing collected packages: python-nmap
Successfully installed python-nmap-0.7.1

Successfully installed python-nmap-0.7.1

[notice] A new release of pip is available: 25.2 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
[notice] To update, run: python.exe -m pip install --upgrade pip
(venv) PS C:\Users\gunav\OneDrive\Desktop\GreenGANProject> python nmap_automation.py
C:\Users\gunav\OneDrive\Desktop\GreenGANProject\venv\Scripts\python.exe: can't open file 'C:\Users\gunav\OneDrive\Desktop\GreenGANProject\venv\Scripts\python.exe': [Errno 2] No such file or directory
(venv) PS C:\Users\gunav\OneDrive\Desktop\GreenGANProject> cd C:\Users\gunav\Downloads
(venv) PS C:\Users\gunav\Downloads> python nmap_automation.py
C:\Users\gunav\OneDrive\Desktop\GreenGANProject\venv\Scripts\python.exe: can't open file 'C:\Users\gunav\OneDrive\Desktop\GreenGANProject\venv\Scripts\python.exe': [Errno 2] No such file or directory
(venv) PS C:\Users\gunav\OneDrive\Desktop\GreenGANProject> cd C:\Users\gunav\Downloads
(venv) PS C:\Users\gunav\Downloads> python nmap_automation.py
C:\Users\gunav\OneDrive\Desktop\GreenGANProject\venv\Scripts\python.exe: can't open file 'C:\Users\gunav\OneDrive\Desktop\GreenGANProject\venv\Scripts\python.exe': [Errno 2] No such file or directory
(venv) PS C:\Users\gunav\Downloads> python nmap_automation.py
Enter target IP: 127.0.0.1
Starting scan...
Starting scan...
Scan complete. Report saved to scan_report.txt
(venv) PS C:\Users\gunav\Downloads>
```

A screenshot of a Notepad++ text editor window. The window has a dark theme and a menu bar with File, Edit, and View. The title bar shows the file name 'scan\_'. The text content is a network scan report. It starts with 'Network Scan Report', followed by 'Timestamp: 2026-03-16 17:02:43.696132' and 'Target: 127.0.0.1'. Under the heading 'Open Ports:', it lists 'Host: 127.0.0.1', 'Port: 135 | Service: msrpc | Version:', and 'Port: 445 | Service: microsoft-ds | Version:'. The report concludes with 'Scan Completed Successfully.' and a cursor on the next line. The status bar at the bottom indicates 'Ln 12, Col 1', '221 characters', 'Plain text', '100%', 'Windows (CRLF)', and 'UTF-8'.

```
Network Scan Report
Timestamp: 2026-03-16 17:02:43.696132
Target: 127.0.0.1

Open Ports:

Host: 127.0.0.1
Port: 135 | Service: msrpc | Version:
Port: 445 | Service: microsoft-ds | Version:

Scan Completed Successfully.
|
```

Port 135 – MSRPC

Used for Windows Remote Procedure Calls.

Can be targeted for lateral movement attacks.

Port 445 – SMB

Used for file sharing.

Historically exploited by attacks like WannaCry.